Channel representations of noisy coherent detection in continuous-variable quantum key distribution: Additive-noise versus attenuator

A. S. Naumchik,[1, *] Roman Goncharov,[1, 2, †] and Alexei D. Kiselev[3, 4, ‡]

[1] *Quantum Information Laboratory, ITMO University, Kadetskaya Line, 3/2, St. Petersburg, 199034, Russia*

[2] *SMARTS-Quanttelecom LLC, 6th Line of Vasilyevsky Island, 59/1B, St. Petersburg, 199178, Russia*

[3] *Laboratory for Quantum Communications, ITMO University, Birzhevaya Line, 16, Saint Petersburg 199034, Russia*

[4] *Laboratory of Quantum Processes and Measurements, ITMO University, Kadetskaya Line 3b, Saint Petersburg 199034, Russia*

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We study the security of continuous-variable quantum key distribution (CV-QKD) under asymmetric coherent detection within an untrusted-detector model. We analyze attenuator channel mode and the additive-noise channel. For both homodyne and double-homodyne detection, we derive the corresponding channel parameters and compare the resulting Holevo information and asymptotic secret fraction.

For the parameter sets considered, receiver asymmetry degrades the protocol performance for both channel models, but neither model gives uniformly more or less pessimistic security estimates. For homodyne detection, the attenuator model gives a lower secret fraction at high transmittance but retains a positive secret fraction over a longer distance.

In the double-homodyne case, the channel parameters and the Holevo information, and hence the resulting secret fraction, depend on the squeezing parameter r that defines ideal measurements.For the parameter sets considered, the Holevo information as a function of r is concave for the additive-noise channel and convex for the attenuator channel. It follows that both the choice of r and the channel representation affect the resulting secret fraction.

I. INTRODUCTION

Homodyne and double-homodyne (heterodyne) detection are standard coherent-detection schemes in quantum optics and continious-variable (CV) quantum information [1–3]. Homodyne detection implements a Gaussian measurement of a single field quadrature, whereas double-homodyne detection performs a joint but necessarily noisy measurement of two conjugate quadratures. These schemes find wide application across various domains, ranging from quantum state tomography [4, 5] and quantum state preparation [6] to a broad class of quantum communication protocols, including continuous-variable quantum key distribution (CV-QKD) [7–9].

In the limit of a strong local oscillator, assuming an ideal (noiseless) setup, these coherent measurement schemes implement projective measurements onto quadrature eigenstates for homodyne detection and coherent states for double homodyne detection. Under these conditions, they function as Gaussian measurements, yielding a Gaussian probability distribution of outcomes for any input Gaussian state [10–12].

A practical coherent receiver is charactarized by a wide range of imperfections, such as non-unit photodetector efficiencies, electronic noise, beam-splitter imbalance, and mismatched responses of the detection branches [13–17]. Consequently, imperfect coherent measurements are naturally described by positive operator-valued measures (POVMs). Measurement imperfections give rise to noisy POVMs [12, 18, 19].

To describe these imperfect measurements, the formalism of quantum channels is employed. Since quantum channels generally describe the evolution of open quantum systems, they provide a natural framework for modeling the physical degradation of information during the measurement process [18, 20–22]. Formally, a quantum channel $\mathcal{C}$ is a linear, completely positive, and trace-preserving (CPTP) map that transforms an input density matrix into an output state. In this framework, the statistics of noisy measurements can be represented either by applying a CPTP channel to the incoming state prior to an ideal measurement, or, equivalently, by applying the dual map $\mathcal{E}^*$ to the ideal POVM [12, 18].

Channel representation is particularly useful in CV-QKD because it incorporates receiver imperfections into the same formalism used to model quantum signal propagation and to evaluate the information leaked to an eavesdropper.

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* Email address: naumchik95@gmail.com
† Email address: rkgoncharov@itmo.ru
‡ Email address: alexei.d.kiselev@gmail.com

To this end, various channel models are available [23], with the most prominent being the attenuator channel [24] and additive-noise channel [25]. The former models the interaction by mixing the input mode with an environmental mode at a beam splitter, whereas the latter introduces a zero-mean Gaussian random displacement whose quadratures are uncorrelated with the input state. With suitable channel parameters, the two channels reproduce the same conditional outcome distributions after the outcome rescaling.

Equivalence at the level of measurement statistics does not, however, imply equivalence as a quantum measurement process. A POVM fixes the outcome probabilities but does not by itself fix the corresponding quantum information transferred to unobserved degrees of freedom [20, 22, 26]. Using a particular channel representation in the security analysis introduces an additional physical assumption whose interpretation depends on whether receiver noise is trusted. In a trusted-detector model, the degrees of freedom associated with calibrated receiver noise remain inside Bob's laboratory [27–31]. In the untrusted-detector model adopted here, the environmental output of the selected channel is assigned to Eve. Different channels may then give the same calibrated receiver statistics and the same mutual information while producing different Alice-Bob states, different environmental outputs, and different values of the Holevo information. More recent works refine this dichotomy by considering several levels of trust for loss and noise and by separating Eve's knowledge of detector noise from her ability to control it [32–34]. Purification of a fixed state is related by an isometry on the purifying system [35]. Here, applying the alternative channels to the incoming state produces different Alice-Bob states before the ideal measurement. Selecting a particular channel representation therefore introduces a physical assumption about the coupling of the optical signal to inaccessible degrees of freedom.

Representing calibrated detector noise as an effective loss has previously been used in trusted and protocol-specific receiver models [36–39]. Such equivalence at the level of receiver statistics does not by itself determine whether the corresponding environmental mode is accessible to Eve.

In the present work, we construct and analyze an attenuator channel of the imperfect homodyne and double-homodyne POVMs and compare it with the additive-noise channel. The channel parameters are obtained by requiring the dual channel to map the ideal POVM onto the target noisy POVM. For homodyne detection, the environmental mode is chosen to be in the vacuum state, so the attenuator takes the form of a pure-loss channel. For double-homodyne detection, the generally anisotropic measurement noise is reproduced using a pure Gaussian environmental state with unequal quadrature variances.We compare the two channels within the Gaussian-modulated coherent-state (GMCS) protocol [40] with reverse reconciliation in the asymptotic regime and evaluate the asymptotic secret fraction under collective Gaussian attacks [41, 42]. In the security model adopted here, receiver asymmetry is treated as untrusted; accordingly, the environmental output of the selected channel is assigned to Eve.

This paper is organized as follows. In Sec. II, we briefly discuss the formalism behind POVMs of imperfect homodyne and double homodyne measurement schemes. In Sec. III, we model these noisy coherent detection schemes using the attenuator and additive-noise quantum channels. In Sec. IV, we calculate informational quantities defining asymptotic secret fraction in the GMCS protocol for both channel models and both coherent-measurement schemes and compare the results. Finally, in Sec. V, we provide a discussion of the results as well as concluding remarks.

II. IMPERFECT GAUSSIAN MEASUREMENTS

We begin with a brief discussion of asymmetrical coherent measurements, schematically depicted in Fig. 1. Asymmetry effects are induced by imbalance (unequal transmission and reflection coefficients) of the beamsplitters and unequal quantum efficiency of the detectors. For detailed derivations of the expressions in this section see Ref. [25].

Q symbol of the POVM describing noisy homodyne measurements (see Fig. 1a) is given by

$$\mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_G}} \exp\left[-\frac{(x-\langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \tag{1}$$

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \phi = \arg\alpha_L, \tag{2}$$

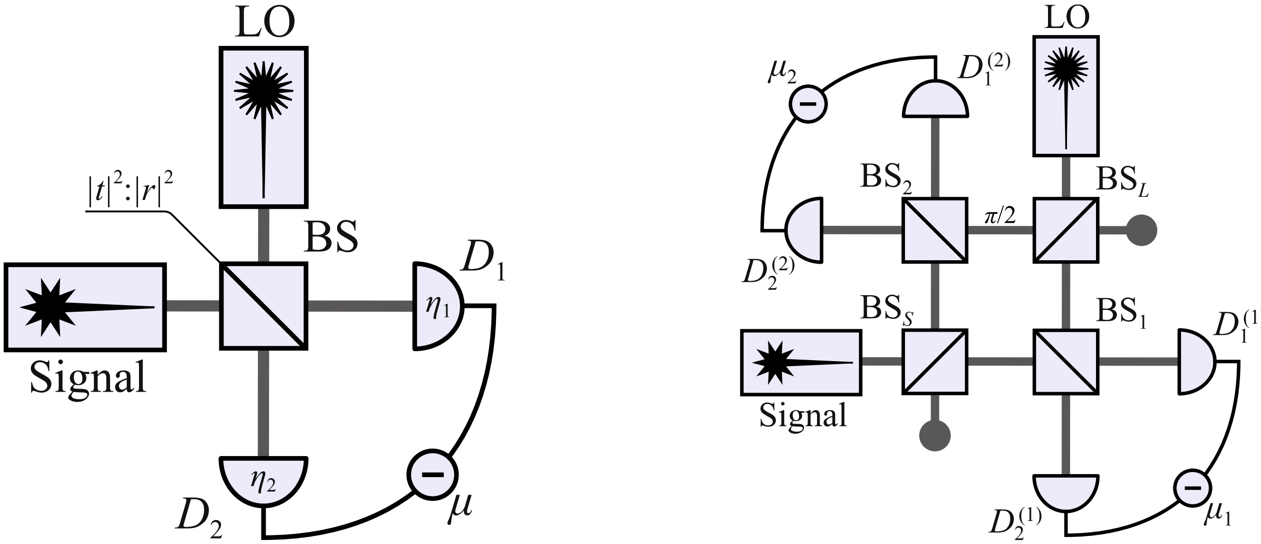
$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \tag{3}$$

where α (α_L) is the complex amplitude of signal (LO) mode, $\hat{x}_\phi$ is the phase-rotated quadrature operator; x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1+\eta_2)CS\,|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1+\eta_2)\,CS}\,|\alpha_L| \tag{4}$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1+\eta_2)CS]^2}. \tag{5}$$



(a) (b)

Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beamsplitter with transmission and reflection coefficients C and S, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double-homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

By computing inverse Forier transform of Q symbol (1), we obtain anti-normal ordered characteristic function [12]. From this characteristic function we can deduce the covariance matrix of the noisy homodyne measurements as follows

$$\Sigma_m^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r} + \sigma_N, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{6}$$

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{7}$$

where $\Sigma_{\mathrm{id}}^{(\mathrm{H})}$ is the covariance matrix of ideal homodyne measurement, $R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$ is the rotation matrix; σ_N is the excess noise variance that accounts for asymmetry effects is given by

$$\sigma_N = \sigma_x - 1. \tag{8}$$

For double homodyne detection, Q symbol is given by

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \exp\left[-\frac{(x_1 - \mathrm{Re}\,\alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \mathrm{Im}\,\alpha e^{-i\phi})^2}{\sigma_2}\right], \tag{9}$$

$$\sigma_G^{(i)} = (\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2)|\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L\alpha_L, \quad \alpha_L^{(2)} = -iS_L\alpha_L, \tag{10}$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)})C_1 S_1 C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)} S_1^2 - \eta_2^{(1)} C_1^2)|\alpha_L^{(1)}| \right\}, \tag{11}$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)})C_2 S_2 S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)} S_2^2 - \eta_2^{(2)} C_2^2)|\alpha_L^{(2)}| \right\}, \tag{12}$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{13}$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2} \tag{14}$$

give the quadrature variances σ_1 and σ_2.

Following the same procedure used to derive Eq. (6), we obtain the covariance matrix of the double homodyne POVM,

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1, \delta_2) R^{\mathrm{T}}(\phi), \tag{15}$$

$$\delta_i = 2\sigma_i - 1. \tag{16}$$

The POVM that describes ideal double-homodyne measurements uses the set of squeezed states,

$$|z, r, \phi\rangle = \hat{R}(\phi)\hat{D}(z)\hat{S}(r)|0\rangle = \hat{D}(ze^{i\phi})\hat{S}(re^{2i\phi})|0\rangle, \tag{17}$$

where z is the coherent state amplitude, r is real-valued squeezing parameter, and ϕ is squeezing phase, $\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}$ is the displacement operator, and $\hat{S}(r) = e^{\frac{1}{2}\left(r\hat{a}^{\dagger 2} - r^*\hat{a}^2\right)}$ is the squeezing operator.

The covariance matrix of squeezed states reads

$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^{\mathrm{T}}(\phi). \tag{18}$$

An important point is that, for the difference between covariance matrices of noisy and ideal double-homodyne measurements, given by Eq. (15) and Eq. (18) respectively, is positive semi-definite, the squeezing parameter r must be ranged in the interval defined by parameters of the scheme (see Ref. [25])

$$r \in [r_2, r_1]\,,\ r_1 = \ln \delta_1^{1/2},\ r_2 = \ln \delta_2^{-1/2}. \tag{19}$$

III. GAUSSIAN CHANNELS MODELING IMPERFECT MEASUREMENTS

For arbitrary state $\hat{\rho}$ and POVM $\hat{\Pi}^{(x)}$, the outcome probability is [35]

$$p(x|\hat{\rho}) = \mathrm{Tr}\left(\hat{\rho}\,\hat{\Pi}^{(x)}\right). \tag{20}$$

The measurement statistics of Gaussian states with noisy Gaussian measurements can be calculated by applying the noise map $\mathcal{C}$ to ρ_G, or the dual map $\mathcal{C}^*$ to the ideal POVM $\hat{\Pi}_{\mathrm{id}}$ [12, 23], i.e.

$$p(x|\hat{\rho}_G) = \mathrm{Tr}\left\{\mathcal{C}(\hat{\rho}_G)\,\hat{\Pi}_{\mathrm{id}}^{(x)}\right\} = \mathrm{Tr}\left\{\hat{\rho}_G\,\mathcal{C}^*\left(\hat{\Pi}_{\mathrm{id}}^{(x)}\right)\right\}. \tag{21}$$

An important property is that the same measurement statistics can be described by different Gaussian channels $\mathcal{C}$, provided their dual maps $\mathcal{C}^*$ yield the same covariance matrix. While the first moments will differ, it is not as crucial due to being able to rescale them.

Note that, as measurements are specified to be Gaussian, the channels $\mathcal{C}$ and $\mathcal{C}^*$ will be Gaussian as well. It follows that the channel $\mathcal{C}$ is fully characterized by matrices $X_{\mathcal{C}}$ and $Y_{\mathcal{C}}$, transforming the first and second moments of Gaussian state ρ_G, $\boldsymbol{r}_G$ and Σ_G respectively, as follows [43]

$$\begin{aligned} \boldsymbol{r}_G &\mapsto X_{\mathcal{C}}\boldsymbol{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}}\Sigma_G X_{\mathcal{C}}^{\mathrm{T}} + Y_{\mathcal{C}}. \end{aligned} \tag{22}$$

The dual channel $\mathcal{C}^*$ transforming the first and second moments as follows

$$\begin{aligned} \boldsymbol{r}_G &\mapsto X_{\mathcal{C}}^{-1}\boldsymbol{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}}^{-1}\Sigma\left(X_{\mathcal{C}}^{-1}\right)^{\mathrm{T}} + X_{\mathcal{C}}^{-1}Y_{\mathcal{C}}\left(X_{\mathcal{C}}^{-1}\right)^{\mathrm{T}}. \end{aligned} \tag{23}$$

In what follows, we will consider two distinct quantum channels to describe the imperfect measurements, namely additive-noise channel and attenuator channel. In accordance with Eq. (21), channel $\mathcal{C}$ will be defined in a way that its' dual channel $\mathcal{C}^*$ will transform the covariance matrix of ideal measurement $\Sigma_{\rm id}$ into the covariance matrix of noisy Gaussian measurement Σ_m,

$$\mathcal{C}^*: \quad \Sigma_{\rm id}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} \tag{24}$$

A. Additive-noise channel

First, we consider additive-noise (classical mixing) channel [12] Φ_γ, characterized by matrices

$$X_\Phi^{(\gamma)} = \mathbb{1}, \quad Y_\Phi^{(\gamma)} \geq 0, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{25}$$

This is a self-dual channel, $\Phi_\gamma^* = \Phi_\gamma$, that does not affect the first moment and changes the second moment as follows

$$\Phi_\gamma : \Sigma \mapsto \Sigma + Y_\Phi^{(\gamma)}. \tag{26}$$

To find matrices $Y_\Phi^{(\gamma)}$ so that (see Eq. (24))

$$\Phi_\gamma : \quad \Sigma_{\rm id}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \Sigma_{\rm id}^{(\gamma)} + Y_\Phi^{(\gamma)}, \tag{27}$$

we need to find the difference between covariance matrices of noisy and ideal measurements, i.e.

$$Y_\Phi^{(\gamma)} = \Sigma_m^{(\gamma)} - \Sigma_{\rm id}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \tag{28}$$

where we defined the covariance matrix of excess noise $\Sigma_N^{(\gamma)}$.

For homodyne detection, from Eq. (7) and (6), we have

$$\Sigma_N^{(\mathrm{H})} = \Sigma_m^{(\mathrm{H})} - \Sigma_{\rm id}^{(\mathrm{H})} = R(\phi)\mathrm{diag}(\sigma_N, 0)R^{\mathrm{T}}(\phi) \geq 0, \tag{29}$$

where σ_N is given by Eq. (8).

For double homodyne detection, from Eq. (15) and (18), we have

$$\Sigma_N^{(\mathrm{DH})}(r) = \Sigma_m^{(\mathrm{DH})} - \Sigma_{\rm id}^{(\mathrm{DH})} = R(\phi)\begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^{\mathrm{T}}(\phi) \geq 0, \tag{30}$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \tag{31}$$

Note that the elements of $\Sigma_N^{(\mathrm{DH})}$ are dependent on squeezing parameter that lies in the interval (19). When $r \in [r_2, r_1]$, it is certain that the covariance matrix of excess noise is positive semi-definite, $\Sigma_N^{(\mathrm{DH})}(r) \geq 0$.

B. Attenuator channel

We shall move on to the attenuator channel $\mathcal{E}_\gamma$, characterized by matrices [12]

$$X_\mathcal{E}^{(\gamma)} = \cos\theta_\gamma \mathbb{1}, \quad Y_\mathcal{E}^{(\gamma)} = \sin^2\theta_\gamma \Sigma_{\rm env}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \tag{32}$$

where $\Sigma_{\rm env}^{(\gamma)}$ is the covariance matrix of the state decribing the enviroment.

Using Eq. (24) together with Eq. (22), we construct the dual channel $\mathcal{E}_N^{*(\gamma)}$ so that it transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing the covariance matrix $\Sigma_{\rm id}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows

$$\mathcal{E}_\gamma^* : \quad \Sigma_{\rm id}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta_\gamma}\Sigma_{\rm id}^{(\gamma)} + \tan^2\theta_\gamma \Sigma_{\rm env}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{33}$$

First, we will consider homodyne detection. If the enviroment is decribed by vacuum, i.e.

$$\Sigma_{\text{env}}^{(\text{H})} = \mathbb{1}, \tag{34}$$

together with from matrices of ideal (7) and noisy (6) measurements, Eq. (33) takes the form

$$\Sigma_m^{(\text{H})} = \lim_{r\to+\infty} R(\phi)\,\text{diag}(e^{-2r} + \sigma_N, e^{2r})\,R^{\text{T}}(\phi) = \lim_{r\to+\infty}\left\{\frac{1}{\cos^2\theta_{\text{H}}} R(\phi)\,\text{diag}(e^{-2r}, e^{2r})\,R^{\text{T}}(\phi) + \tan^2\theta_{\text{H}}\mathbb{1}\right\} \tag{35}$$

due to the limit $r \to \infty$, we may write

$$\frac{1}{\cos^2\theta}\Sigma_{\text{id}}^{(\text{H})} \sim \Sigma_{\text{id}}^{(\text{H})}, \quad e^{2r} + \text{const} \sim e^{2r}, \tag{36}$$

it is sufficient to solve for $\tan^2\theta_{\text{H}}$, i.e.

$$\tan^2\theta_{\text{H}} = \sigma_N, \tag{37}$$

where σ_N is defined by Eq. (8). Then, $\cos^2\theta_{\text{H}}$ and $\sin^2\theta_{\text{H}}$ are given by

$$\cos^2\theta_{\text{H}} = \frac{1}{\sigma_x}, \quad \sin^2\theta_{\text{H}} = \frac{\sigma_N}{\sigma_x}. \tag{38}$$

For double-homodyne detection with $\delta_1 \neq \delta_2$, the environmental state obtained below is generally anisotropic for an arbitrary admissible value of r, although it can be isotropic at a special value of r. In the attenuator channel model studied here, we restrict the environmental mode to a pure state with $\det \Sigma_{\text{env}}^{(\text{DH})} = 1$,

$$\Sigma_{\text{env}}^{(\text{DH})} = R(\phi)\,\text{diag}\left(a, \frac{1}{a}\right)\,R^{\text{T}}(\phi), \quad a \geq 0. \tag{39}$$

Solving Eq. (33) for the case of double homodyne detection, (see Eq. (18), (15)),

$$\Sigma_m^{(\text{DH})} = R(\phi)\,\text{diag}(\delta_1, \delta_2) R^{\text{T}}(\phi) = R(\phi)\left\{\frac{1}{\cos^2\theta_{\text{DH}}}\,\text{diag}(e^{2r}, e^{-2r}) + \tan^2\theta_{\text{DH}}\,\text{diag}\left(a, \frac{1}{a}\right)\right\} R^{\text{T}}(\phi), \tag{40}$$

$$\begin{cases} \delta_1 = \frac{e^{2r}}{\cos^2\theta_{\text{DH}}} + \tan^2\theta_{\text{DH}} a \\ \delta_2 = \frac{e^{-2r}}{\cos^2\theta_{\text{DH}}} + \tan^2\theta_{\text{DH}} a^{-1} \end{cases}. \tag{41}$$

Solving the system yields

$$\cos^2\theta_{\text{DH}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)} = \frac{\delta_1 e^{-2r} + \delta_2 e^{2r} - 2}{\delta_1\delta_2 - 1}, \tag{42}$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r} - 1}. \tag{43}$$

Note that at the ends of the r-interval (19), $r \to r_2$ or $r \to r_1$, either $a(r)$ or $1/a(r)$ tend to infinity, making the ancilla state (39) the eigenstate of the quadrature operator (3). By evaluating in this case the equations (41), i.e. resolving indeterminate form $\tan^2\theta_{\text{DH}} a(r)$ for $r = r_2$ and $\tan^2\theta_{\text{DH}}/a(r)$ for $r = r_1$, because one of noise variances (31) would be 0, it can be seen that the effect of this channel would be the same as the effect of the additive-noise channel (27) modeling double homodyne detection with the same r. This fact will be used in numerical calculations (see next section and Fig. 3).

IV. GAUSSIAN-MODULATED COHERENT-STATE PROTOCOL

In the Gaussian-modulated coherent-state (GMCS) protocol, Alice samples the complex amplitude $\alpha = \alpha_1 + i\alpha_2$ from the Gaussian prior:

$$p_A(\alpha) = \frac{1}{\pi V_{\text{A}}}\exp\left[-\frac{|\alpha|^2}{2V_{\text{A}}}\right]. \tag{44}$$

She then prepares the coherent state $|\alpha\rangle$. The states pass the composite Gaussian channel $\mathcal{N}_{A\to B} = \mathcal{L}_N \circ \mathcal{L}_{T_{\rm ch}}$, where index indicates pure loss channel transmission coefficient, $T_{\rm ch}$ is the channel transmittance, $N = 1+\xi_{\rm ch}/(1-T_{\rm ch})$, and $\xi_{\rm ch}$ is the channel excess noise. Then, the receiver uses either homodyne or double-homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the following expressions (see Ref. [25] for derivation)

$$I_{\rm AB}^{(\rm H)} = \frac{1}{2}\log\left[1+\frac{4T_{\rm ch}V_{\rm A}}{\sigma_x+\xi_{\rm ch}}\right], \tag{45}$$

$$I_{\rm AB}^{(\rm DH)} = \frac{1}{2}\sum_i \log\left[1+\frac{4T_{\rm ch}V_{\rm A}}{2\sigma_i+\xi_{\rm ch}}\right], \tag{46}$$

where σ_x and σ_i given by Eqs. (5) and (14), respectively, and upper indices hereinafter stand for homodyne and double-homodyne detection.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel reads (see Ref. [29])

$$\mathcal{I}_A\otimes\mathcal{L}_N\circ\mathcal{L}_{T_{\rm ch}}: \quad \Sigma_{\rm TMSV} = \begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z \\ \sqrt{V^2-1}\sigma_z & V\mathbb{1}\end{pmatrix} \mapsto \Sigma_{\rm AB} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1}\end{pmatrix}, \tag{47}$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix $\Sigma_{\rm AB}$ are given by

$$V = 1+4V_{\rm A}, \quad c = \sqrt{T_{\rm ch}(V^2-1)}, \tag{48}$$

$$V_{\rm B} = T_{\rm ch}(V-1)+1+\xi_{\rm ch}. \tag{49}$$

We consider local action of either attenuator channel $\mathcal{E}_\gamma$ (32) or additive-noise channel Φ_γ (25), on second mode of TMSVS shared by Alice and Bob (47). In both cases, mutual information is given by $I_{\rm AB}^{(\gamma)}$ (45), (46), as it is classical.

It follows

$$\mathcal{I}_A\otimes\mathcal{E}_\gamma : \Sigma_{\rm AB}\mapsto\Sigma_{\rm AB}^{(\mathcal{E}_\gamma)} = \begin{pmatrix} V\mathbb{1} & c\cos\theta_\gamma\sigma_z \\ c\cos\theta_\gamma\sigma_z & \cos^2\theta_\gamma\left(V_{\rm B}+\tan^2\theta_\gamma\Sigma_{\rm env}^{(\gamma)}\right)\end{pmatrix}, \tag{50}$$

$$\mathcal{I}_A\otimes\Phi_N^{(\gamma)} : \Sigma_{\rm AB}\mapsto\Sigma_{\rm AB}^{(\Phi_\gamma)} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1}+\Sigma_N^{(\gamma)}\end{pmatrix}, \quad \gamma\in\{\rm H, DH\}. \tag{51}$$

When the channel $\mathcal{C}_\gamma\in\{\Phi_\gamma,\mathcal{E}_\gamma\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat\rho_{\rm AB}^{(\mathcal{C}_\gamma)} = \mathcal{I}_A\otimes\mathcal{C}_\gamma(\hat\rho_{\rm AB})$ and the measurements performed by the receiver are assumed to be noiseless. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\rm EB}^{(\mathcal{C}_\gamma)} = S_{\rm E}-S_{\rm E|B} = S_{\rm AB}^{(\mathcal{C}_\gamma)} - S_{\rm A|B}^{(\mathcal{C}_\gamma)}, \quad \mathcal{C}_\gamma\in\{\Phi_\gamma,\mathcal{E}_\gamma\}. \tag{52}$$

gives the Holevo information. The joint entropy

$$S_{\rm AB}^{(\mathcal{C}_\gamma)} = g(\nu_1^{(\mathcal{C}_\gamma)})+g(\nu_2^{(\mathcal{C}_\gamma)}), \tag{53}$$

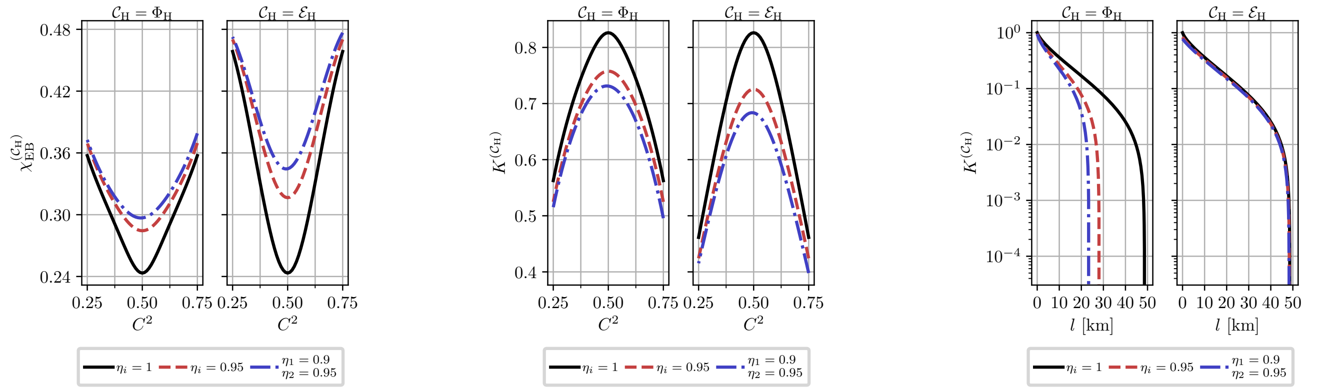
$$g(\nu) = \frac{\nu+1}{2}\log\frac{\nu+1}{2} - \frac{\nu-1}{2}\log\frac{\nu-1}{2}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (51) given by

$$\nu_{1,2}^{(\mathcal{C}_\gamma)} = \sqrt{\Delta_{\mathcal{C}_\gamma}/2\pm\sqrt{\Delta_{\mathcal{C}_\gamma}^2/4-D_{\mathcal{C}_\gamma}}}, \quad D_{\mathcal{C}_\gamma} = \det\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}, \tag{54}$$

$$\Delta_{\mathcal{C}_\gamma} = \det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{11}) + \det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{22}) + 2\det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{12}), \tag{55}$$

where $[\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{ii}$ ($[\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}$.



(a) (b) (c)

Figure 2: (a) Holevo information, $\chi_{\mathrm{EB}}^{(C_{\mathrm{H}})}$ and (b) asymptotic secret fraction, $K^{(C_{\mathrm{H}})}$, as a function of the beam splitter transmission, C^2, of the homodyne receiver computed at different values of the detector efficiencies for additive-noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{\mathrm{H}})}$ versus transmission length l for additive-noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

Conditional entropies $S_{\mathrm{E|B}} = S_{\mathrm{A|B}}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{\mathrm{A|B}}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{\mathrm{A|B}}^{(\gamma)}}, \tag{56}$$

$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)})^{-1}\sigma_z. \tag{57}$$

Direct computations show that these conditional matrices and therefore their symplectic eigenvalues are independent on channel model. Symplectic eigenvalues are given by

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \tag{58}$$

$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{(V_{\mathrm{B}} + \delta_1)(V_{\mathrm{B}} + \delta_2)}}, \tag{59}$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [44]

$$K^{(\mathcal{C}_\gamma)} = \beta I_{AB}^{(\mathcal{C}_\gamma)} - \chi_{EB}^{(\mathcal{C}_\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \tag{60}$$

As symplectic eigenvalues for additive-noise channel are calculated per formulas Eq. (54) and do not allow for simplification, in the following sections we focus only on analytic expressions for symplectic eigenvalues for attenuator channel model.

A. Homodyne detection

For attenuator channel, from Eq. (50) and (38) we have

$$\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}(V_{\mathrm{B}} + \sigma_N)\,\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{61}$$

$$\tilde{V}_{\mathrm{B}} \equiv T(V-1) + 1 + \xi, \tag{62}$$

$$\tilde{c} \equiv \sqrt{T(V^2-1)}, \tag{63}$$

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_{\mathrm{B}}$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ,

$$T = T_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{64a}$$

$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}. \tag{64b}$$

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (64a) and channel noise (64b) by a factor of σ_x^{-1}.

Symplectic eigenvalues of $\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})}$, $\nu_{1,2}^{(\mathcal{E}_{\mathrm{H}})}$, can be computed using formulas [43]:

$$\nu_{1,2}^{(\mathcal{E}_{\mathrm{H}})} = \frac{z \pm [\tilde{V}_{\mathrm{B}} - V]}{2}, \quad z = \sqrt{\left(V + \tilde{V}_{\mathrm{B}}\right)^2 - 4\tilde{c}^2}, \tag{65}$$

and then substitued in Eq. (53) to calculate joint entropy $S_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (52) and asymptotic secret fraction (60) versus beam splitter transmission C^2, as well as secret fraction versus transmission length, for both additive-noise (51) and attenuator (50) channels. While channel noise is reduced in this setup (64b), so is channel transmission (64a), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (45), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive-noise channel.

B. Double-homodyne detection

For attenuator channel, from Eq. (50) and (42) we have

$$\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{DH}})} = \begin{pmatrix} V\mathbb{1} & & c\cos\theta_{\mathrm{DH}}\sigma_z & \\ & & & \\ c\cos\theta_{\mathrm{DH}}\sigma_z & & \cos^2\theta_{\mathrm{DH}}\left(V_{\mathrm{B}} + \delta_1 - \frac{e^{2r}}{\cos^2\theta_{\mathrm{DH}}}\right) & 0 \\ & & 0 & \cos^2\theta_{\mathrm{DH}}\left(V_{\mathrm{B}} + \delta_2 - \frac{e^{-2r}}{\cos^2\theta_{\mathrm{DH}}}\right) \end{pmatrix} \tag{66}$$

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (54). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (19), like in the case of additive-noise channel (see Ref. [25]).

Fig. 3 illustrates dependence of Holevo information on the squeezing parameter, for both additive-noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double-homodyne detection with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [25]), with equal values of Holevo information for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of Holevo information on squeezing parameter is symmetrical around $r = 0$ for both channels. For additive-noise channel, $r = 0$ is the point of maximal Holevo information, while for attenuator channel $r = 0$ corresponds to minimal Holevo information. Moreover, while Holevo information is unidefined at $r = r_{2,1}$ (see Eq. (42)), the values tend to the same values of joint entropy in the case of additive-noise channel, as signified by color-matched white-filled dots in Fig. 3.

For the asymmetric parameter sets considered, where $\delta_1 \neq \delta_2$, the Holevo information for both channel models becomes asymmetric about its extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For these parameter sets, the Holevo information is concave in r for the additive-noise channel and convex in r for the attenuator channel.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive-noise (51) and attenuator (50) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maximum of joint entropy for both models is either at $r = r_1$ or $r = r_2$. Moreover, the upper bound for additive-noise channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (52) and asymptotic secret fraction (60) on the signal beam splitter transmission C_S^2, as well as secret fraction versus transmission length if the receiver uses double-homodyne

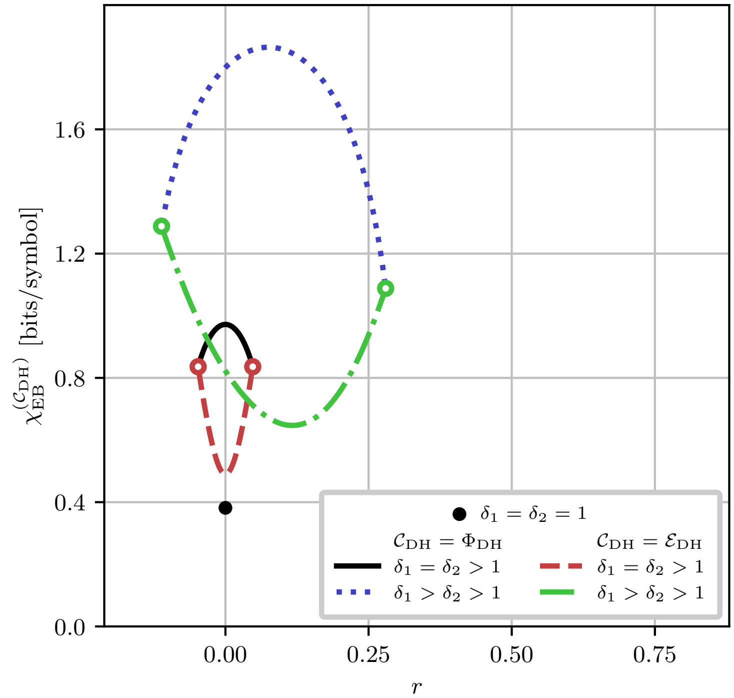


Figure 3: Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{DH};\mathcal{C}_{\mathrm{DH}})}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dashdotted curves correspond to additive-noise channel model, while dashed and dotted curves are for attenuator channel model. Color-matched white-filled dots mark points where Holevo information is undefined at the boundaries of the r interval (19) (see Eq. (43) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; and $\xi_{\mathrm{ch}} = 10^{-2}$.

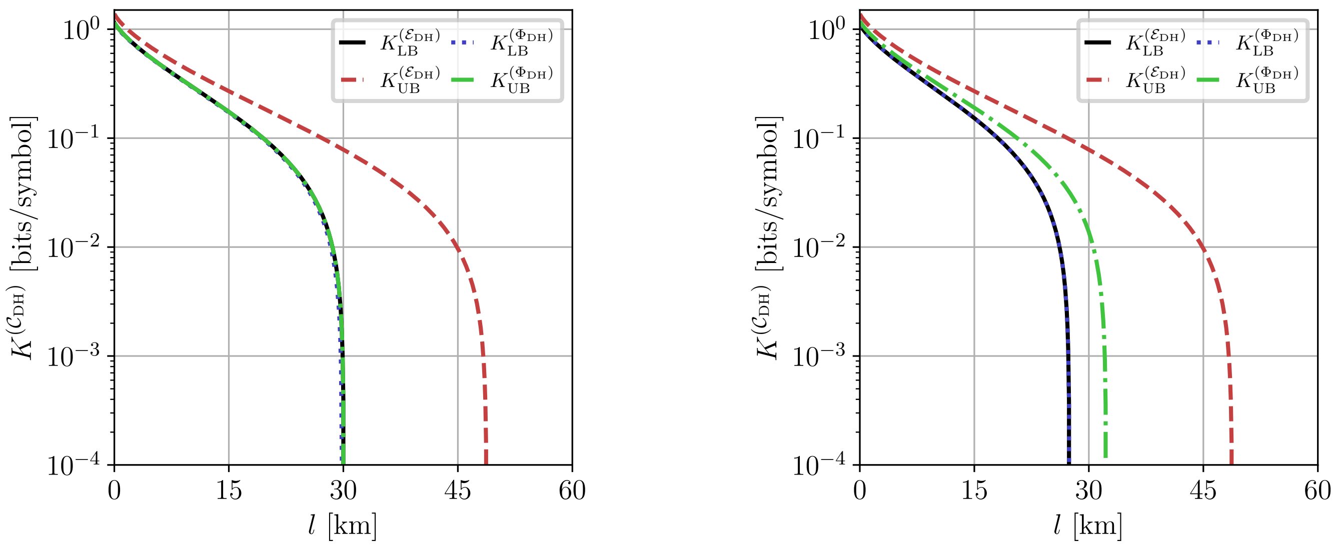
detection, for both additive-noise (51) and attenuator (50) channels. Unlike the homodyne detection case (see Fig. 2), Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where ideal measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2+r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive-noise channel.

Analogously to the case of homodyne detection (see Fig. 2c), in the case of double-homodyne detection the attenuator channel has greater reach than the additive-noise channel, as illustrated by Fig. 5c.

V. DISCUSSION AND CONCLUSION

In this paper, we have constructed and compared additive-noise and attenuator channel representations of noisy homodyne and double-homodyne measurements affected by receiver asymmetry. Our approach requires the dual of the selected Gaussian channel, together with a deterministic invertible rescaling of the measurement outcomes, to reproduce the target noisy Gaussian measurement. At the level of covariance matrices, this condition is expressed by Eq. (24). These conditions constrain the channel parameters; under the assumptions adopted here, the parameters are then determined by the receiver parameters and, for double-homodyne detection, by r.

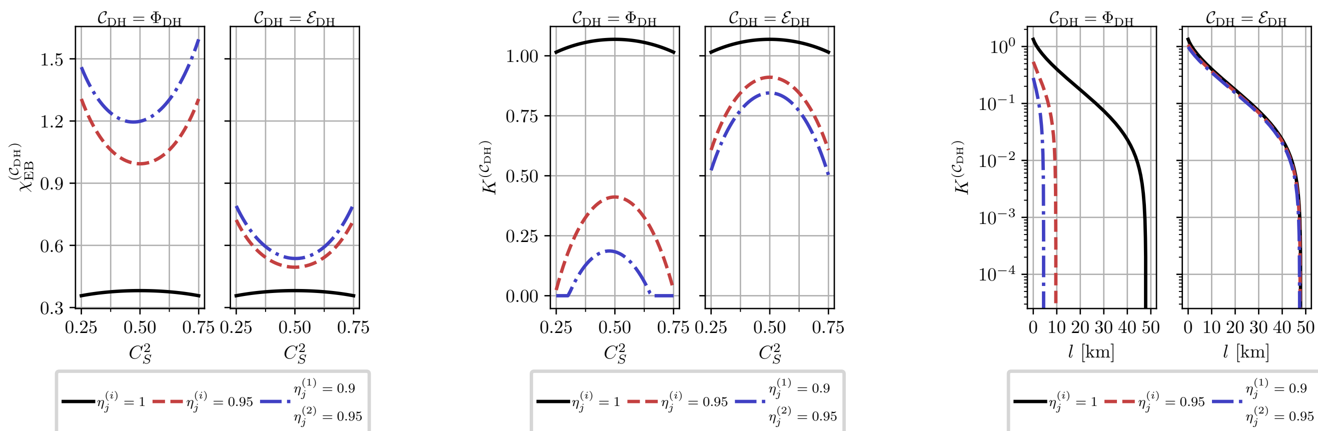
After the deterministic invertible rescaling of the measurement outcomes, the two channels reproduce the same conditional statistics of Bob's outcomes. They therefore give the same Alice-Bob mutual information. For fixed source and channel parameters, the conditional covariance matrix in Eq. (57), and hence the conditional entropy $S_{A|B}^{(\gamma)}$, are the same for the two channels because the detector imperfections enter the conditional state through the same noisy measurement. The difference between the security estimates originates from the joint state of Alice and Bob. The additive-noise and attenuator channels generally produce different joint covariance matrices and, consequently,



(a) $q = 1$

(b) $q = 1.25$

Figure 4: Asymptotic secret fraction $K^{(C_{\rm DH})}$ versus channel length l, computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(14)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive-noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\rm ch} = 10^{-2}$; losses are 20 dB per 100 km.



(a)

(b)

(c)

Figure 5: (a) Holevo information, $\chi_{\rm EB}^{(C_{\rm DH})}$ and (b) asymptotic secret fraction, $K^{(C_{\rm DH})}$, as a function of the signal beam splitter transmission, C_S^2, of the double-homodyne receiver computed at different values of the detector efficiencies for additive-noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5$; $T_{\rm ch} = 0.95$; $\xi_{\rm ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{\rm DH})}$ versus transmission length l for additive-noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\rm ch} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel models $r = \frac{r_2+r_1}{2}$ (see main text).

different joint entropies. In the untrusted-detector model considered in this paper, the environmental output of the selected channel is assigned to Eve. The corresponding Holevo information (52) therefore depends on the selected channel even when Bob's observed statistics remain unchanged.

For homodyne detection, we take the environmental mode of the attenuator channel to be in the vacuum state, so that the detector imperfections are represented by a pure-loss channel. This channel rescales both the effective transmission and the output-referred excess noise by the same factor, as shown in Eq. (64).

The numerical results in Fig. 2 show that the additive-noise and attenuator channels give significantly different security estimates. At high channel transmittance and for the parameters considered, the attenuator channel gives a larger Holevo information and a lower asymptotic secret fraction. At longer distances, however, the secret fraction remains positive up to a greater channel length than in the additive-noise channel. Therefore, neither channel can be described as strictly more or less pessimistic.

A distinguishing feature of double-homodyne detection is the nonuniqueness of the representation of its noisy POVM. The covariance matrix of the physical double-homodyne measurement is fixed by the receiver parameters and does not depend on the squeezing parameter r. The effects of the ideal Gaussian POVM are proportional to projectors onto displaced squeezed states and depend on r. The noisy POVM is obtained by applying the dual of the selected Gaussian channel to the ideal POVM, together with the outcome rescaling. The requirement that the excess-noise covariance matrix be positive semidefinite restricts r to the interval given by Eq. (19).

For the additive-noise channel, variation of r redistributes the measurement noise between the two conjugate quadratures according to Eq. (31). For the attenuator channel, the same variation changes both the attenuation coefficient and the covariance matrix of the environmental state, as described by Eqs. (42) and (43). In the attenuator channel considered here, the generally anisotropic measurement noise is reproduced by an environmental mode in an anisotropic Gaussian state, which was chosen to be pure.

At the boundaries of the admissible interval of the squeezing parameter r, one of the additive-noise quadrature variances vanishes. In the attenuator channel, the environmental state approaches an infinitely squeezed limit: one quadrature variance tends to zero, whereas the conjugate variance diverges. The boundary points are therefore understood as limiting channels in which the attenuator result approaches the corresponding additive-noise result.

Figure 3 shows that the Holevo information depends on r in qualitatively different ways for the two channel models. For the parameter sets considered, this dependence is concave for the additive-noise channel and convex for the attenuator channel. When the measurement noise is symmetric between the two quadratures, both curves are symmetric about $r = 0$. In this case, the coherent state, given by $r = 0$, maximizes the Holevo information for the additive-noise channel and minimizes it for the attenuator channel. For the asymmetric case considered, this symmetry is lost and the extrema are shifted away from $r = 0$.

The lower and upper secret-fraction curves in Fig. 4 describe the range obtained by varying r over its admissible interval for each channel model. For the near-ideal quadrature variances used in Fig. 4, the lower bounds of the two channels nearly coincide because the Holevo-information maximum occurs at, or close to, the interval boundaries. The upper bounds become more clearly separated when the imbalance of the signal beam splitter is increased.

The comparison shown in Fig. 5 was performed at the midpoint of the admissible interval of r. In the symmetric case, this value coincides with the maximum of the additive-noise Holevo information and the minimum of the attenuator Holevo information. The pronounced separation between the secret fractions in Fig. 5 therefore reflects both the selected channel model and the selected squeezing parameter.

These results clarify the role of the chosen channel representation when receiver noise is treated as untrusted. Although the additive-noise and attenuator channels reproduce the same receiver statistics after outcome rescaling, they generally yield different joint states of Alice and Bob and different environmental outputs assigned to Eve. Choosing a particular channel representation therefore introduces a physical assumption about the coupling between the optical signal and the inaccessible degrees of freedom. A security estimate should not be based on a particular channel representation solely because it gives a larger secret fraction. The selected representation should either be justified from the physical structure of the receiver or the dependence of the estimate on the channels included in the analysis should be reported explicitly. The present comparison is restricted to the additive-noise channel and to attenuator channels whose environmental modes are in pure Gaussian states. Mixed environmental states and other Gaussian channel representations compatible with the same receiver statistics were not considered.

Appendix A extends the receiver model to include independent pure losses at the output ports of a lossless beam splitter. These losses can be incorporated into the effective photodetector efficiencies. By contrast, a general lossy asymmetric beam splitter with unequal complex transmission and reflection amplitudes does not, in general, preserve the simple quadrature form of the receiver response and cannot be included by merely replacing the detector efficiencies.

The security analysis was performed in the asymptotic limit for the GMCS CV-QKD protocol under collective Gaussian attacks, with the modeled receiver imperfections treated within an untrusted-detector model. Trusted receiver noise, finite-size corrections, active detector attacks, and uncertainties in the calibration parameters require separate treatment. Within these assumptions, equivalent receiver statistics can lead to different estimates of Eve's information when different channel representations are used within the untrusted-detector model.

ACKNOWLEDGEMENTS

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Appendix A: Lossy beam splitter

In this appendix, we show that applying independent pure loss channels at the output ports of a lossless beam splitter yields the exact same transformation as a single formally defined lossy beam splitter.

First, consider the case of homodyne detection. The lossless unbalanced beam splitter mixes the input signal mode $\hat{a}$ and the local oscillator mode $\hat{a}_L$. This transformation is described by the unitary rotation matrix $\mathbf{U}_{\mathrm{BS}}$,

$$\begin{aligned}\begin{pmatrix}\hat{a}_1\\ \hat{a}_2\end{pmatrix} &= \mathbf{U}_{\mathrm{BS}}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} = \begin{pmatrix}C & S\\ -S & C\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix}\\ &= \begin{pmatrix}C\hat{a}+S\hat{a}_L\\ -S\hat{a}+C\hat{a}_L\end{pmatrix},\end{aligned} \tag{A1}$$

where C and S are the transmission and reflection coefficients of the beam splitter BS, respectively.

We model the subsequent losses by passing each intermediate output mode $\hat{a}_{1,2}$ through a virtual beam splitter of transmittance $\tau_{1,2}$. This mixes each signal with an independent environmental vacuum mode $\hat{v}_{1,2}$, where $[\hat{v}_i, \hat{v}_j^\dagger] = \delta_{ij}$,

$$\begin{pmatrix}\hat{b}_1\\ \hat{c}_1\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1} & -\sqrt{1-\tau_1}\\ \sqrt{1-\tau_1} & \sqrt{\tau_1}\end{pmatrix}\begin{pmatrix}\hat{a}_1\\ \hat{v}_1\end{pmatrix}, \tag{A2}$$

$$\begin{pmatrix}\hat{b}_2\\ \hat{c}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_2} & -\sqrt{1-\tau_2}\\ \sqrt{1-\tau_2} & \sqrt{\tau_2}\end{pmatrix}\begin{pmatrix}\hat{a}_2\\ \hat{v}_2\end{pmatrix}. \tag{A3}$$

Tracing out the lost environmental modes $\hat{c}_{1,2}$, the remaining physical output modes $\hat{b}_{1,2}$ expand to

$$\begin{pmatrix}\hat{b}_1\\ \hat{b}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1}C & \sqrt{\tau_1}S\\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} + \begin{pmatrix}-\sqrt{1-\tau_1}\hat{v}_1\\ -\sqrt{1-\tau_2}\hat{v}_2\end{pmatrix}. \tag{A4}$$

From Eq. (A4), we extract the effective scattering matrix $\mathbf{T}_{\mathrm{BS}}$ and the fluctuation noise operators $\hat{F}_{1,2}$ representing the device excitations

$$\mathbf{T}_{\mathrm{BS}} = \begin{pmatrix}\sqrt{\tau_1}C & \sqrt{\tau_1}S\\ -\sqrt{\tau_2}S & \sqrt{\tau_2}C\end{pmatrix}. \tag{A5}$$

It is not difficult to show that to preserve the Canonical Commutation Relations for the output fields, $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$, the noise operators $\hat{F}_{1,2}$ must satisfy

$$[\hat{F}_i, \hat{F}_j^\dagger] = \delta_{ij} - \sum_k T_{ik}T_{jk}^*, \tag{A6}$$

where T_{ik} are the elements of the scattering matrix $\mathbf{T}_{\mathrm{BS}}$ (A5). Evaluating this constraint explicitly using the elements of $\mathbf{T}_{\mathrm{BS}}$ yields:

$$[\hat{F}_1, \hat{F}_1^\dagger] = 1-\tau_1, \tag{A7}$$

$$[\hat{F}_2, \hat{F}_2^\dagger] = 1-\tau_2, \tag{A8}$$

$$[\hat{F}_1, \hat{F}_2^\dagger] = 0, \tag{A9}$$

which matches the commutators calculated from Eq. (A4).

We now connect this setup to the formal, standard mathematical treatment of lossy beam splitters [45, 46]. This framework models loss via an interaction matrix $\mathbf{A}$ acting on ancillary modes, which must obey the positivity condition $\mathbf{1} - \mathbf{A}\mathbf{A}^\dagger \geq 0$ to satisfy unitarity. Note that the extracted scattering matrix $\mathbf{T}_{\mathrm{BS}}$ (A5) can be factored into a product of a diagonal scaling matrix and the lossless unitary,

$$\begin{aligned}\mathbf{T}_{\mathrm{BS}} &= \begin{pmatrix}\sqrt{\tau_1} & 0\\ 0 & \sqrt{\tau_2}\end{pmatrix}\begin{pmatrix}C & S\\ -S & C\end{pmatrix}\\ &= \begin{pmatrix}\sqrt{\tau_1} & 0\\ 0 & \sqrt{\tau_2}\end{pmatrix}\mathbf{U}_{\mathrm{BS}}.\end{aligned} \tag{A10}$$

By defining the positive semi-definite matrix representing the loss properties as $\mathbf{A}\mathbf{A}^\dagger = \mathrm{diag}(1-\tau_1, 1-\tau_2)$, it follows that:

$$\sqrt{\mathbf{1}-\mathbf{A}\mathbf{A}^\dagger} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix}. \tag{A11}$$

Substituting this back into Eq. (A10) yields the polar decomposition of the scattering matrix of a lossy beam splitter in its generalized form [46]:

$$\mathbf{T}_{\mathrm{BS}} = \sqrt{\mathbf{1}-\mathbf{A}\mathbf{A}^\dagger}\,\mathbf{U}_{\mathrm{BS}}. \tag{A12}$$

Because the matrix $\mathbf{A}\mathbf{A}^\dagger$ representing the losses is diagonal, it is straightforward to see that to account for lossy beam splitter in a homodyne setup it is sufficient to simply rescale the quantum efficiencies of detection: $\eta_{1,2} \to \tau_{1,2}\eta_{1,2}$. The same argument would apply in the case of double-homodyne detection.

We conclude this appendix by discussing the generalized (asymmetrical) beam splitter model where input ports of the beam splitter are characterized by different set of transmission and reflection coefficients. In this case, the non-unitary scattering matrix is written in the form [47]

$$\mathbf{S} = \begin{pmatrix} t & \rho \\ r & \tau \end{pmatrix}, \qquad \mathbf{S}\mathbf{S}^\dagger \le \mathbf{1}, \tag{A13}$$

where t, τ (r, ρ) are complex transmission (reflection) coefficients.While it can be factored into lossy and unitary parts similarly to (A12), this case cannot be covered by the formulas after some replacement, because it ruins quadrature representation:

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_1, \alpha_2\rangle, \tag{A14}$$

$$\alpha_1 = t\alpha + \rho\alpha_L, \quad \alpha_2 = r\alpha + \tau\alpha_L, \tag{A15}$$

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{A16}$$

$$\begin{aligned} \sigma_G &= \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \\ &= \left(\eta_1|\rho|^2 + \eta_2|\tau|^2\right)|\alpha_L|^2 + \left(\eta_1|t|^2 + \eta_2|r|^2\right)|\alpha|^2 \\ &\quad + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* + \eta_2 r\tau^*\right)\alpha\alpha_L^*\right] \\ &\approx \left(\eta_1|\rho|^2 + \eta_2|\tau|^2\right)|\alpha_L|^2 + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* + \eta_2 r\tau^*\right)\alpha\alpha_L^*\right], \end{aligned} \tag{A17}$$

$$\begin{aligned} \mu_G &= \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \\ &= \left(\eta_1|\rho|^2 - \eta_2|\tau|^2\right)|\alpha_L|^2 + \left(\eta_1|t|^2 - \eta_2|r|^2\right)|\alpha|^2 \\ &\quad + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* - \eta_2 r\tau^*\right)\alpha\alpha_L^*\right] \\ &\approx \left(\eta_1|\rho|^2 - \eta_2|\tau|^2\right)|\alpha_L|^2 + 2\,\mathrm{Re}\left[\left(\eta_1 t\rho^* - \eta_2 r\tau^*\right)\alpha\alpha_L^*\right]. \end{aligned} \tag{A18}$$

Appendix B: Fluctuating Local Oscillator

In this Appendix, we extend the theoretical treatment of homodyne measurements to account for classical intensity fluctuations of the local oscillator (LO).

We start from the Gaussian approximation of the photon count difference probability distribution in homodyne scheme [25]

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{B1}$$

$$\sigma_G = \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \approx \left(\eta_1 S^2 + \eta_2 C^2\right)|\alpha_L|^2, \tag{B2}$$

$$\mu_G = \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \approx \left(\eta_1 S^2 - \eta_2 C^2\right)|\alpha_L|^2 + CS(\eta_1 + \eta_2)\,|\alpha_L|\,\langle\hat{x}_\phi\rangle, \tag{B3}$$

where

$$G(x;\sigma) \equiv \frac{1}{\sqrt{2\pi\sigma}}\exp\left(-\frac{x^2}{2\sigma}\right). \tag{B4}$$

We assume that the LO amplitude fluctuates around its mean value α_L such that

$$\alpha_L = \alpha_L + \delta\alpha_L, \quad \delta\alpha_L \sim G(\delta\alpha_L, \sigma_L) \tag{B5}$$

where σ_{LO} represents the variance of the LO amplitude fluctuations.

To find the total probability distribution $\mathsf{P}_{\mathrm{tot}}(\mu)$ that accounts for the local oscillator fluctuations, we perform an ensemble average by integrating the conditional distribution $\mathsf{P}_G(\mu)$ over the Gaussian prior of the LO amplitude.

$$\mathsf{P}_{\mathrm{total}}(\mu) = \int \mathrm{d}\delta\alpha_L\, \mathsf{P}_G(\mu|\delta\alpha_L)\mathsf{P}_G(\delta\alpha_L) = \tag{B6}$$

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